

# Thermal Systems

## Thermal resistance and capacitance: the thermometer as a first-order lag.

Ogata, *Modern Control Engineering*, Ch. 4.

In this tutorial you will:

- model a thermometer from a heat balance,
- recognize the familiar first-order time constant  $\tau = RC$ , and
- see how a thicker wall (larger  $R$ ) slows the response.

Step through with **Ctrl+Enter**, or render a report with **publish**.

Thermal systems, like fluid systems, are characterized by a resistance and a capacitance:

- **Thermal resistance**  $R$ : relates heat flow rate  $q$  to temperature difference  $\Delta\theta$  across it,  $R = \Delta\theta/q$ .
- **Thermal capacitance**  $C$ : relates stored heat to temperature,  $C = dQ/d\theta$  (mass times specific heat for a lumped body).

Classic example: a mercury thermometer of capacitance  $C$  immersed in a liquid bath at temperature  $\theta_i(t)$ , with thermal resistance  $R$  between the bath and the thermometer's mercury (its glass wall). The thermometer's heat balance ( $C\dot{\theta}_o = \text{net heat flow in}$ ) gives

$$C\dot{\theta}_o = \frac{\theta_i - \theta_o}{R} \quad \Rightarrow \quad RC\dot{\theta}_o + \theta_o = \theta_i$$

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## Transfer Function of the Thermometer

Taking the Laplace transform with zero initial conditions:

$$(RCs + 1)\Theta_o(s) = \Theta_i(s) \quad \Rightarrow \quad G(s) = \frac{\Theta_o(s)}{\Theta_i(s)} = \frac{1}{RCs + 1}$$

This is again a single first-order lag, exactly analogous to the electrical RC circuit and the fluid tank: a thermometer plunged into a hot bath does not read the new temperature instantly, but approaches it exponentially with time constant  $\tau = RC$ .

With  $R = 0.9^\circ\text{C} \cdot \text{s}/\text{cal}$  and  $C = 0.1 \text{ cal}/^\circ\text{C}$ :

```
R = 0.9; C = 0.1;
tau = R*C;
G_thermo = tf(1,[tau 1])

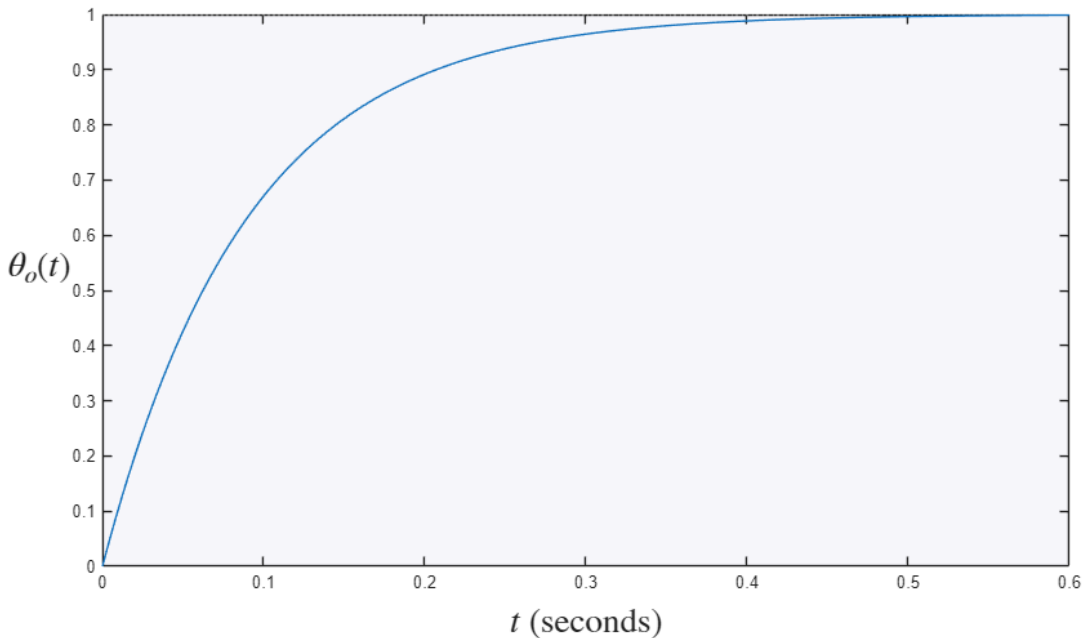
figure
step(G_thermo)
title('Thermometer Step Response','Interpreter','latex','FontSize',20)
ylabel('$\theta_o(t)$','Interpreter','latex','FontSize',20)
set(get(gca, 'YLabel'), 'Rotation', 0)
xlabel('$t$', 'Interpreter','latex','FontSize',20)

G_thermo =

      1
-----
0.09 s + 1

Continuous-time transfer function.
```

## Thermometer Step Response



At  $t = \tau$ , the response has reached  $1 - e^{-1} \approx 63.2\%$  of the final value – consistent with the time-constant rule used for any first-order system in this repository.

```
[y,t] = step(G_thermo);
y_at_tau = interp1(t,y,tau);
disp(['Response at t = tau: ' num2str(y_at_tau) ' (expect ~0.632)'])
```

```
Response at t = tau: 0.63204 (expect ~0.632)
```

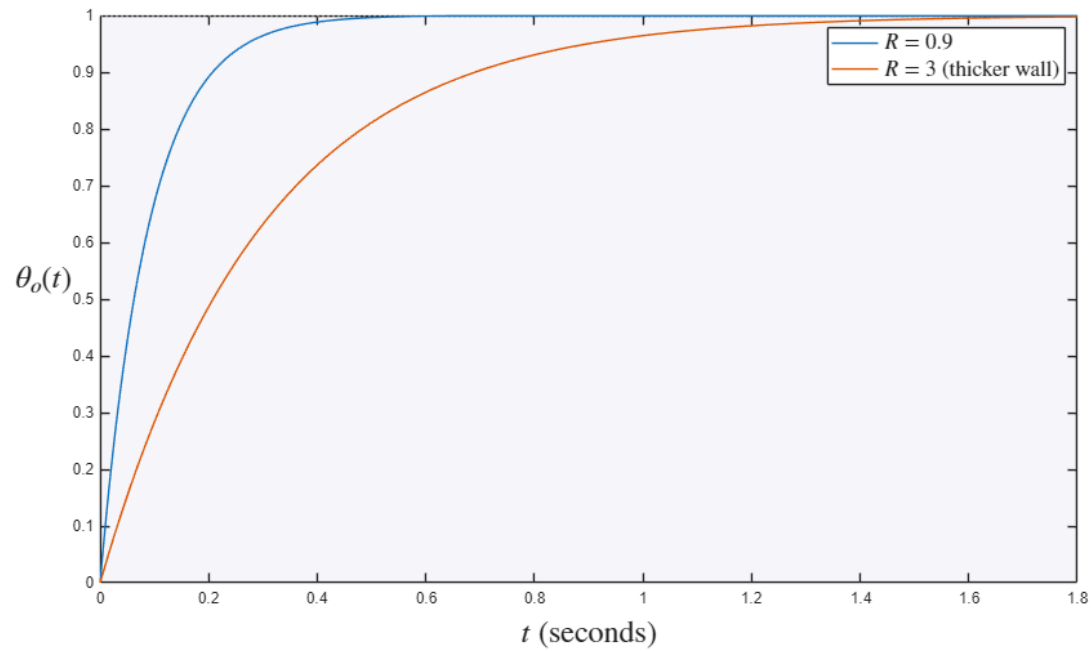
## Effect of Thermometer Wall Thickness (Increasing R)

A thicker glass wall increases  $R$ , slowing the thermometer's response without changing its final reading – a direct illustration of how the time constant, not the DC gain, sets the speed of a first-order thermal system.

```
R_thick = 3;
G_thick = tf(1,[R_thick*C 1]);

figure
hold on
step(G_thermo)
step(G_thick)
hold off
legend('$R=0.9$', '$R=3$ (thicker wall)', 'Interpreter', 'latex', 'FontSize', 14)
title('Effect of Thermal Resistance on Response Speed', 'Interpreter', 'latex', 'FontSize', 20)
ylabel('$\theta_o(t)$', 'Interpreter', 'latex', 'FontSize', 20)
set(get(gca, 'YLabel'), 'Rotation', 0)
xlabel('$t$', 'Interpreter', 'latex', 'FontSize', 20)
```

### Effect of Thermal Resistance on Response Speed



### Try it yourself

- Notice that changing  $R$  moves the time constant but never the final reading – try  $R = 0.3$  and confirm the steady value is unchanged.
- Estimate the time to reach 95% (about  $3 \cdot \tau$ ) and check it on the plot.